

A Prompt Templates

We show the detailed prompt templates for language model evaluation across each domain included in AbsenceBench in Table 7 that are used to obtain the evaluation results in Table 3. Additionally, Table 8 presents prompt templates with adapted instructions used for assessing the NIAH test under the poetry and GitHub PRs domains.

Poetry (default)
System Prompt You are helping a student practice memorizing poems. The student will recite a poem, but they may have missed some lines. Your task is to identify exactly which lines are missing from their recitation. List only the missing lines, nothing else.
User Message Here is the complete original poem: {original poem} Now, here is my recitation which may be missing some lines: {modified poem} What lines did I miss? Please list only the missing lines, nothing else.
Numerical Sequences (default)
System Prompt You are helping a student practice reciting sequences. The student will recite a sequence, but they may have missed some numbers. Your task is to identify exactly which numbers are missing from their recitation. List only the missing numbers, nothing else
User Message Here is a sequence of numbers: {original sequence} Now, here is my recitation of the sequence which may be missing some numbers: {modified sequence} What numbers did I miss? Please list only the missing numbers, nothing else.
GitHub PRs (default)
System Prompt You are helping a software developer determine if their merge of a pull request was successful. The developer had to edit the commit history and just wants to make sure that they have not changed what will be merged. They will list the changed lines. Your job is to figure out if they have missed any insertions or deletions from the original merge. Only pay attention to the insertions and deletions (ignore the context of the diff).
User Message Here is the complete original diff: {original diff} And here is the merge diff after the developer fixed the commit history: {modified diff} What changed lines (insertions or deletions) present in the original diff are missing in the merge diff (if any)? List only the missing changed lines, nothing else.

Table 6: Detailed prompt templates used for evaluating language models under each domain of AbsenceBench

Poetry (<i>post-instruction</i>)
System Prompt You are helping a student practice memorizing poems. The student will recite a poem, but they may have missed some lines. Your task is to identify exactly which lines are missing from their recitation. List only the missing lines, nothing else. User Message {original poem} {modified poem} The original poem is followed by my recitation of the poem which may be missing some lines. What lines did I miss? Please list only the missing lines, nothing else.
Numerical Sequences (<i>post-instruction</i>)
System Prompt You are helping a student practice reciting sequences. The student will recite a sequence, but they may have missed some numbers. Your task is to identify exactly which numbers are missing from their recitation. List only the missing numbers, nothing else User Message {original sequence} {modified sequence} The original sequence is followed by my recitation of the sequence which may be missing some numbers. What numbers did I miss? Please list only the missing numbers, nothing else.
GitHub PRs (<i>post-instruction</i>)
System Prompt You are helping a software developer determine if their merge of a pull request was successful. The developer had to edit the commit history and just wants to make sure that they have not changed what will be merged. They will list the changed lines. Your job is to figure out if they have missed any insertions or deletions from the original merge. Only pay attention to the insertions and deletions (ignore the context of the diff). User Message {original diff} {modified diff} The original diff is followed by a merge diff after the developer fixed the commit history What changed lines (insertions or deletions) present in the original diff are missing in the merge diff (if any)? List only the missing changed lines, nothing else.

Table 7: Perturbed prompt templates where the task instructions are positioned after the original and modified documents.

Poetry (NIAH)
System Prompt You are helping a student practice memorizing poems. The student will recite a poem, but they may have added some random lines that related to Harry Potter characters. Your task is to identify exactly which lines are not in the original poem. List only the missing lines, nothing else. User Message Here is the complete original poem: {original poem} Now, here is my recitation with some extra lines that is related to Harry Potter novel series: {modified poem} What lines did I miss? Please list only the extra lines, nothing else.
GitHub PRs (NIAH)
System Prompt You are helping a software developer determine if their merge of a pull request was successful. The developer had to edit the commit history and accidentally added some random lines related to Harry Potter characters. They will list the changed lines. Your job is to figure out if they have added any insertions from the original merge. Only pay attention to the insertions. User Message Here is the complete original diff: {original diff} And here is the merge diff after the developer fixed the commit history: {modified diff} What changed lines (insertions or deletions) present in the original diff are missing in the merge diff (if any)? List only the missing changed lines, nothing else.

Table 8: Modified prompt templates used for evaluating language models under the NIAH test setting within the Poetry and GitHub PRs domains.

B Inference Details

We evaluate a total of 14 LLMs on AbsenceBench. All model inferences are obtained through API requests. We show the detailed information of each model’s maximum context length, API provider and reference in Table 9.

Models	Context Length	API Provider	API Reference
Gemini-2.5-flash (Google, 2025)	1M	Google	models/gemini-2.5-flash-preview-04-17
Claude-3.7-Sonnet (Anthropic, 2024)	200K	Anthropic	claude-3-7-sonnet-latest
o3-mini (OpenAI, 2025)	200K	OpenAI	o3-mini
GPT-4.1 (OpenAI et al., 2024b)	1M	OpenAI	gpt-4.1
GPT-4.1-mini (OpenAI et al., 2024b)	1M	OpenAI	gpt-4.1-mini
GPT-4o (OpenAI et al., 2024a)	128K	OpenAI	gpt-4o
Grok-3-mini-Beta (xAI, 2025)	131K	xAI	grok-3-mini-beta
Llama-4-Maverick (Meta, 2025)	1M	TogetherAI	meta-llama/Llama-4-Maverick-17B-128E-Instruct-FP8
Llama-3.3-70B-Instruct (Grattafiori et al., 2024)	131K	TogetherAI	meta-llama/Llama-3.3-70B-Instruct-Turbo
Qwen3-235B (Qwen, 2025a)	41K	TogetherAI	Qwen/Qwen3-235B-A22B-fp8-tput
Qwen2.5-72B-Instruct (Yang et al., 2024)	32K	TogetherAI	Qwen/Qwen2.5-72B-Instruct-Turbo
QwQ-32B (Qwen, 2025b)	32K	TogetherAI	Qwen/QwQ-32B
DeepSeek-R1 (DeepSeek-AI et al., 2025)	128K	TogetherAI	deepseek-ai/DeepSeek-R1
Mixtral-8x7B-Instruct (Jiang et al., 2024)	32K	TogetherAI	mistralai/Mixtral-8x7B-Instruct-v0.1

Table 9: Detailed information of models evaluated on AbsenceBench